

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	21 JUNE 2026
Team ID	Individual project
Project Name	Online Complaint Registration and Management System
Maximum Marks	4 Marks

Technical Architecture:

Complaint Management System Technical Architecture

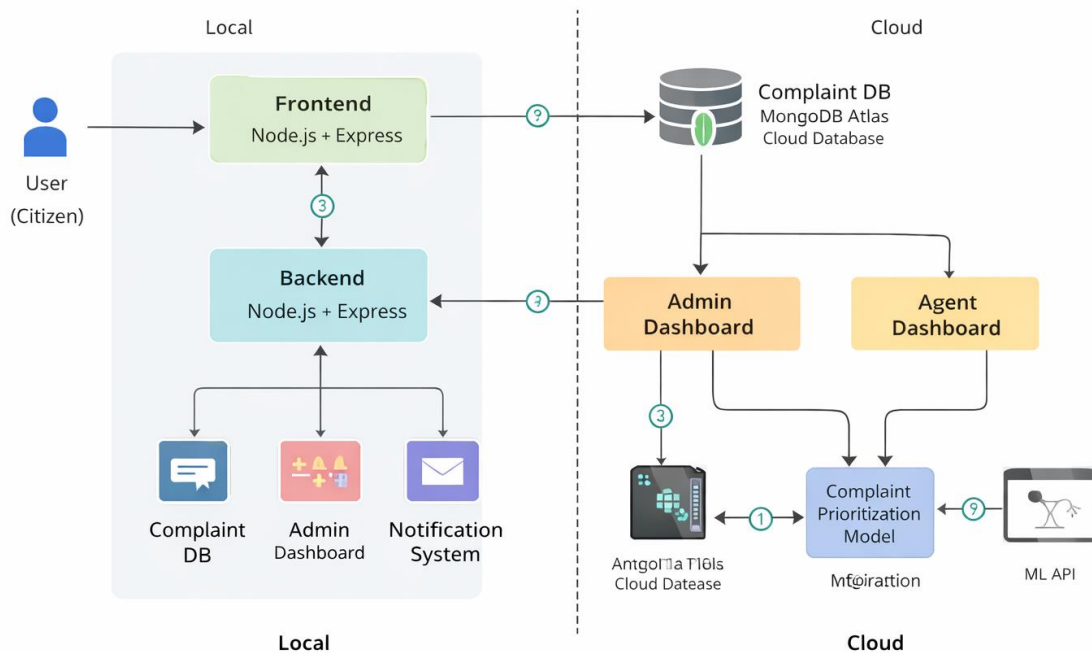


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.		Responsive interface for Users, Admins and Agents	React.js, HTML5, CSS3, Bootstrap
2.	Application Logic-1	User Registration and Login	Node.js, Express.js
3.	Application Logic-2	Complaint Registration and Tracking	Express.js REST API
4.	Application Logic-3	Agent Assignment and Complaint Status Updates	Node.js, Express.js
5.	Database	Stores Users, Complaints, Agents and Feedback	MongoDB, Mongoose
6.	Cloud Database	Cloud-hosted NoSQL database	MongoDB Atlas
7.	File Storage	Stores profile images and project assets (future scope)	Local File System
8.	External API-1	REST API communication between frontend and backend	Fetch API
9.	External API-2	Role-based Authentication and Authorization	JWT, bcrypt.js
10.	Machine Learning Model	Processes client requests and REST APIs	Node.js + Express.js
11.	Infrastructure (Server / Cloud)	Application hosting and deployment	Localhost / Render (Future)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and Backend frameworks used	React.js, Node.js, Express.js
2.	Security Implementations	Secure authentication and password encryption	JWT, bcrypt.js, Protected Routes
3.	Scalable Architecture	Three-tier architecture supporting future scalability	Presentation Layer, Business Layer, Database Layer
4.	Availability	Reliable REST API and cloud database access	MongoDB Atlas, Express Server
5.	Performance	Fast API responses and efficient database queries	REST API, MongoDB Indexing, Mongoose